

Supine Heel Slides

Supine heel slides are performed by laying down on a flat surface and pulling your heel backwards towards your buttocks.

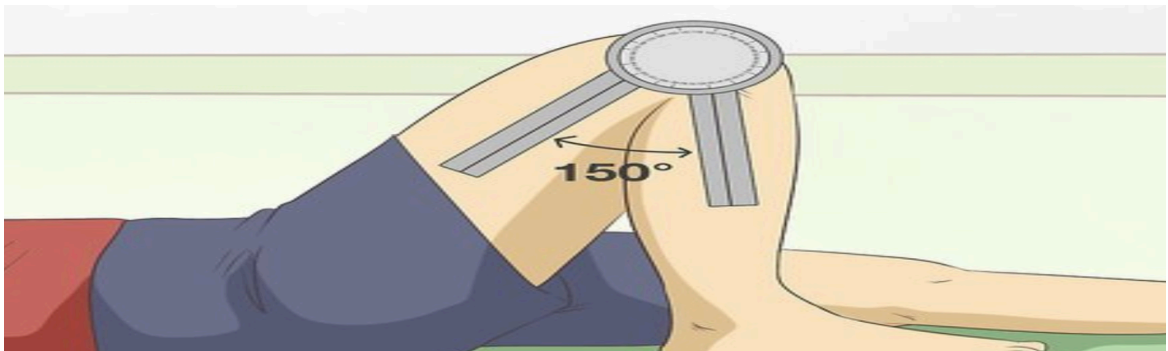
While the knee and hip are fundamentally connected, hip compensation is not ideal during most targeted exercises. Because the body is one continuous kinetic chain, the hip often tries to take over when the targeted muscles are weak or fatigued. However, allowing this compensation defeats the purpose of the exercise, limits your strength gains, and frequently leads to injury.

Challenges with seated heel slides:

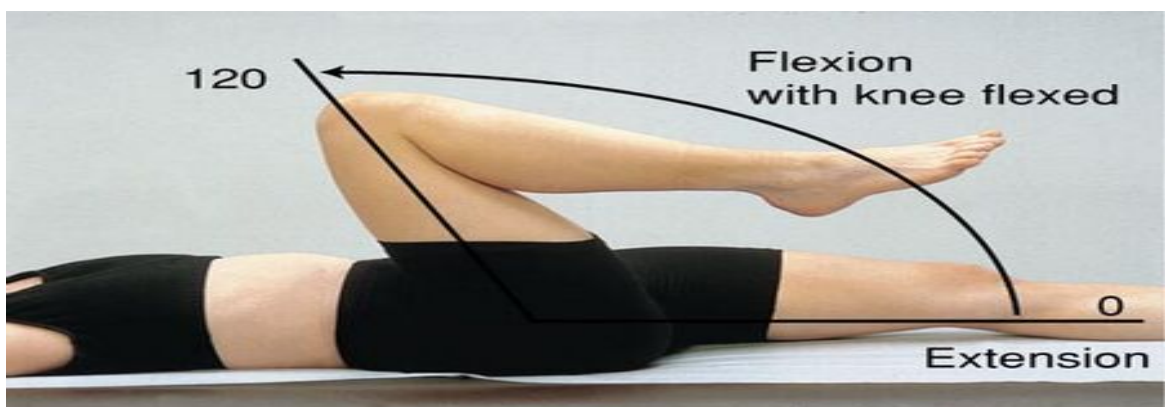
- Trunk compensation and hip stability since the patient has no back support.

Key Angles:

- 1) Knee Flexion Angle (Primary tracking angle)
 - **Knee flexion** is the movement of bending your knee to decrease the angle between your thigh and lower leg. A healthy adult knee typically has a functional active range of motion between 0 degree and 120 degree and 140 degree of flexion.



- 2) Hip Flexion Angle measures the degree to which the thigh is lifted toward the torso, decreasing the angle between the femur and the pelvis. A normal, healthy range of motion for hip flexion is generally between 120° and 140°.



3) Pelvic/ Lower Back Position

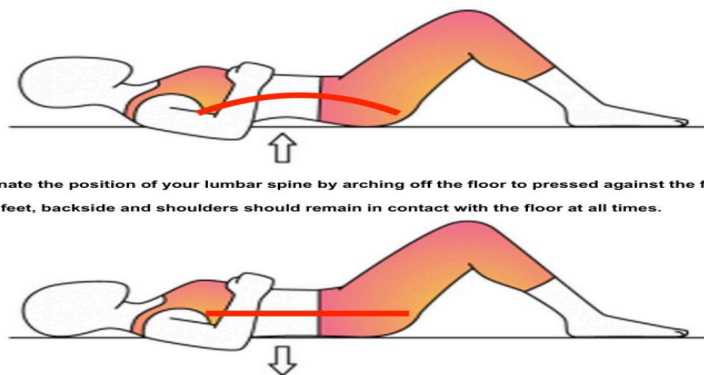
- It is necessary for the back to be firmly pressed against the floor.
- Lower back arching or lifting off the surface is not suitable and is called the **anterior pelvic tilt** or lumbar hypertension. This is one of the most common compensation errors which should be avoided.
- Another major form of compensation is the **pelvic hiking sideways**. This means that one side of the hip hitches upward toward the armpit, making the waist uneven. Instead of the pelvis staying perfectly flat & level, the hip of the moving leg lifts up.

- Why does this error occur?

Ans: When the knee joint is stiff & refuses to bend, the body looks for an easier way to pull the foot backward. By hiking the hip, the patient artificially shortens the length of his/ her leg. This allows the foot to move closer to the body without forcing the knee joint itself to actually bend any further.

This error tricks the patient into thinking his/ her knee flexibility is improving when it is actually staying exactly the same. It overworks a lower back muscle called the quadratus lumborum which can quickly lead to lower back pain or muscle spasms.

- **Anterior Pelvic Tilt happens** because your **core and glutes are weak**, or your hip flexors are too tight. Your body defaults to arching your lower back to cheat its way through an exercise.
- **Pelvic Hiking happens** because your **knee or hip lacks the flexibility to bend or clear the floor**. Your body hitches the hip upward to artificially shorten the leg so you can pull or swing it forward.



Alternate the position of your lumbar spine by arching off the floor to pressed against the floor. Your feet, backside and shoulders should remain in contact with the floor at all times.

4) Resting Leg Stability

- In supine heel slides, only one leg moves. The other leg should remain completely still & flat.
- Any drop compared to the baseline - the patient is compensating with the other leg.

5) Ankle angle

- The ankle angle refers to the **position of your foot relative to your shin**, specifically whether your foot is pointed away from you (**plantarflexion**) or pulled up toward your knee (**dorsiflexion**).
- During supine heel slides, your ankle angle is incredibly important because it changes how smoothly your leg moves, regulates nerve tension, and dictates which muscles in your leg are doing the work.

1) Dorsiflexion

- It engages the front shin muscles (tibialis anterior), which helps "turn off" overactive calf muscles via a neurological process called reciprocal inhibition. It also stabilizes the foot, keeping your leg sliding in a perfectly straight line without rolling outward.
- **The Drawback:** Flexing the ankle stretches the sciatic nerve and the calf muscles. If your leg is highly sensitive or inflamed, pulling the toes up can create uncomfortable nerve tension or resistance behind the knee.

2) Plantarflexion

- It relaxes the calf muscles and slackens the sciatic nerve. If you are experiencing nerve pain, burning, or intense tightness in the back of your leg, pointing your toes removes that tension, making it easier to bend the knee.
- **The Drawback:** It can cause the calf and hamstring muscles to cramp up more easily. It also makes it easier for your foot to flop sideways, which causes the knee to twist as it bends.

For heel slides - we will be focusing on Dorsiflexion.

Summary of Differences		
Feature 	Supine (Lying down)	Seated (On a chair)
Best For	Early post-op, acute pain, initial range	Mid-to-late rehab, breaking plateaus
Hip Position	Extended (flat)	Flexed (90 degrees)
Muscle Effort	Low (passive or active-assisted)	Higher hamstring activation
Common Compensation	Hiking the pelvis sideways	Leaning back or lifting the thigh

Muscle Activation

- Hamstrings are the primary muscles that do the work to pull the heel toward the buttock during a supine heel slide.
- The quadriceps are used in the second half of the exercise to straighten the leg back out.